🟢 Foundation Level — Online Lesson Programme

> *A 5-week introductory programme for students aged 11 and above. No prior coding experience required.*

📢 Dear Parents — Why This Matters

We live in a world being reshaped by Artificial Intelligence. It is already deciding which videos your child watches, which job applications get reviewed first, and how doctors detect diseases. Understanding AI is no longer a specialist skill — it is becoming as fundamental as reading and writing.

This programme does not aim to turn your child into a programmer overnight. Instead, it gives them something more valuable: the confidence to engage with AI tools, the vocabulary to think and talk about them critically, and the creative experience of building something with them.

In these 5 sessions, your child will:

- ✅ Write their first lines of code — through music and dance(no typing anxiety — it starts with blocks)
- ✅ Understand how AI recognises patterns, learns from data, and makes decisions
- ✅ Use Claude, one of the world's leading AI assistants, to build a real, playable game from scratch
- ✅ Learn what a loop, an array, and collision detection are — core computer science ideas — by *seeing* them work inside a program they built themselves
- ✅ Develop prompt engineering skills: the ability to give clear, specific instructions to an AI to get useful results

Why does this matter for your child's future?

Whether your child becomes an engineer, a doctor, an artist, or an entrepreneur — they will work alongside AI. Students who understand how it works, not just how

to use it, will have a lasting advantage. This programme builds that foundation.

What will your child walk away with?

- A fully playable **AI Trivia Quiz** game, built by them using AI
- A customised **Snake game** with features they chose and added
- A **Code.org certificate** of completion
- The knowledge and language to describe what AI is, how it learns, and why it matters

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📋 Programme at a Glance

	Detail
Level	Foundation
Duration	5 weeks · 1 session per week · 1 hour per session
Schedule	Every Saturday · **7:00–8:00pm UK time**
Format	Online (Real-time Hands-on Remote Guidance)
Language	English
Group size	Maximum **8 students** — every student gets individual attention
Age	Minimum 11 years old · No upper limit
Price	**£45 per student** for the full 5-week block
What's included	All lesson materials · Teacher support · Code.org access (free)
What you need	Laptop or desktop · Stable internet · Free Claude.ai account

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📅 Session-by-Session Plan

Session 1 — Welcome to Coding & AI *(1 hour)*

> **Big question:** *What does coding actually feel like — and what does it have to do with music?*

What we do	A short, friendly introduction to the course and what AI really is. Then straight into action: students compose music with loops and program animated dancers with events. No typing — just thinking.
Key concepts	What is AI · Loops · Events · Sequences
Student takeaway	"I wrote my first program and I made something creative with it."

Session 2 — Move to the Beat *(1 hour)*

> **Big question:** *How do you tell a computer to do things in exactly the right order — and what happens when you get it wrong?*

What we do	Students bring together music and dance from Session 1 — this time programming an AI robot to move and perform. The focus is on **sequencing**: putting instructions in the correct order so the robot performs exactly as intended. Students discover what happens when steps are in the wrong order, and why sequence is one of the most fundamental ideas in all of computing. Then they explore more to understand how AI finds patterns in massive datasets.
Key concepts	Sequencing · Machine Learning · Big Data · Pattern recognition
Student takeaway	"The order of instructions matters — and that's true for every program ever written."

Session 3 — Meet Claude · Build a Quiz Game *(1 hour)*

> **Big question:** *Can I describe a game in plain English and have AI build it for me — in under a minute?*

What we do	Introduction to Claude and what Artifacts are. Students use a structured prompt to generate a fully playable AI Trivia Quiz game — with animations, score tracking, and instant feedback — directly in the browser. Then they customise it with their own questions.

Key concepts	What is Claude · Prompt Engineering · Artifacts · HTML · CSS · JavaScript (introduced, not required)
Platform	Claude.ai
Student takeaway	"I built and personalised a real game using AI — just by describing what I wanted."

Session 4 — Snake Game · Build It *(1 hour)*

> **Big question:** *How does Snake actually work — and what does it have in common with GPS and self-driving cars?*

What we do	Before building, students learn the computer science behind Snake: how an array tracks the snake's body, how a game loop runs 10+ times per second, and how collision detection knows when the game is over. Then they generate a complete, fully featured Snake game using Claude Artifacts and test it against a checklist.
Key concepts	Game Loop · Arrays · Collision Detection · Canvas API
Platform	Claude.ai
Student takeaway	"I understand what's happening *inside* the game — not just how to play it."

Session 5 — Snake Game · Improve It + What's Next *(1 hour)*

> **Big question:** *How do I make this game my own — and what could I build next?*

What we do	Students pick two improvements from a feature menu (e.g. Portal Mode, particle explosions, a leaderboard, an AI auto-play mode) and apply them using prompts. Then they do a Code Treasure Hunt — finding key concepts like `snake.push`, `setInterval`, and `localStorage` inside the real source code. The session ends with a reflection and a preview of Studio Level.
Key concepts	Iterative improvement · Reading code · Pathfinding (intro) · localStorage

Platform	Claude.ai
Student takeaway	"I improved my own game and I can actually read parts of the code behind it."

💷 Pricing

Full 5-week Foundation block	**£45 per student**
Payment covers	All 5 sessions · All teaching materials · Teacher support throughout
Not included	Nothing — all accounts are set up together in class

> **Note:** All account registrations (including Claude.ai) are done together as a class with teacher guidance. No preparation needed from parents.

✅ Requirements Checklist

Before the first session, please confirm your child has the following:

☐ **Laptop or desktop computer** — tablets and phones are not suitable (keyboard required for Snake)

☐ **Stable internet connection** — video call + browser tools running simultaneously

☐ **Free access to these websites** — Claude.ai · Code.org · Google AI Quests
(students in regions where these sites are restricted will not be able to participate)

☐ **English reading ability** — all lessons are taught and written in English

🎓 What Comes Next — Studio Level

Students who complete Foundation Level are ready to apply for **Studio Level 1** — where they build a real, deployed web application (a music sampler) using React, and publish it live on the internet with a real URL they can share.

Students who have not completed Foundation Level may still apply for Studio Level — the teacher will conduct a short evaluation to confirm readiness.

Evaluation covers:

- Basic understanding of if/else and loops
- Familiarity with AI and Big Data concepts
- Ability to write a clear prompt to achieve a goal (e.g. generate an image, create a plan)